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from google.colab import files
uploaded = files.upload()

import cv2
import matplotlib.pyplot as plt

img = cv2.imread(list(uploaded.keys())[0])

img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

h, w = img_rgb.shape[:2]
center = (w // 2, h // 2)

M_clockwise = cv2.getRotationMatrix2D(center, -45, 1.0)
clockwise_img = cv2.warpAffine(img_rgb, M_clockwise, (w, h))

M_counter = cv2.getRotationMatrix2D(center, 45, 1.0)
counter_img = cv2.warpAffine(img_rgb, M_counter, (w, h))

plt.figure(figsize=(15,5))

plt.subplot(1,3,1)
plt.title("Original Image")
plt.imshow(img_rgb)
plt.axis('off')

plt.subplot(1,3,2)
plt.title("Clockwise Rotation")
plt.imshow(clockwise_img)
plt.axis('off')

plt.subplot(1,3,3)
plt.title("Counter-Clockwise Rotation")
plt.imshow(counter_img)
plt.axis('off')

plt.show()
```

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Original Image



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Clockwise Rotation



Counter-Clockwise Rotation

